

Report
Of
**DEVELOPMENT OF AN LLM-BASED
INTELLIGENT KNOWLEDGE
RETRIEVAL SYSTEM**

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By
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CANDIDATE'S DECLARATION

I, **Naitik Jain**, do hereby declare that the internship report titled “**Development of an LLM-Based Intelligent Knowledge Retrieval System**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

The report presents the work, technical concepts, and learning undertaken by me during my summer internship at the **Defence Research and Development Organisation (DRDO)** in the Cybersecurity domain.

All references and sources used in preparing this report have been appropriately acknowledged. I affirm that this report represents my own academic work and has not been submitted to any other institute for the fulfilment of any degree or diploma requirement.

I further understand that I shall be solely responsible for any copyright violation in this regard.

Signature

Student Name: Naitik Jain

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ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported me throughout my Summer Internship Program.

I am thankful to the **Defence Research and Development Organisation (DRDO)** for providing me with the opportunity to undertake my internship in the Cybersecurity domain and gain practical exposure to emerging technologies in Artificial Intelligence and Large Language Models.

I am grateful to the professionals and team members who provided guidance and valuable technical insights throughout the internship. Their guidance helped me develop an understanding of Large Language Models, AI agents, knowledge retrieval systems, and various supporting technologies.

I would also like to express my gratitude to **IILM University, Greater Noida**, for providing me with the opportunity to undertake this internship as part of my academic curriculum.

During the internship, I gained exposure to technologies and concepts including **Python, SQL, Virtual Machines, OpenCode, C++, Artificial Intelligence, AI agents, Large Language Models, and knowledge-based information retrieval.**

The internship helped me improve my technical understanding, problem-solving abilities, debugging skills, and ability to work with unfamiliar technologies.

Finally, I would like to thank my parents for their continuous support, encouragement, and blessings throughout my academic journey.

Signature

Student Name: Naitik Jain

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INTERNSHIP COMPLETION CERTIFICATE

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SUBJECT: CERTIFICATE OF INTERNSHIP

This is to certify that **"Mr. Naitik Jain"**, a student of B.Tech in Computer Science & Engineering, **Roll No. 2410030570** from IILM University, Greater Noida has completed his internship at the O/o Advisor (Cyber), DRDO, New Delhi from 10th June 2026 to 07th August 2026. During the period of the internship, He has worked on **"AI Assisted Cyber Security workflow"**.

During the Internship period the candidate was found to be sincere, hardworking, disciplined and dedicated.



(Pankaj Kumar)
Head HR & Scientist 'F'
O/o Advisor Cyber
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4. PROJECT DESCRIPTION

4.1 INTRODUCTION

Artificial Intelligence has emerged as one of the most important areas of modern Computer Science. The development of advanced machine learning techniques and Large Language Models has enabled software systems to understand, process, and generate natural language.

Large Language Models, commonly referred to as LLMs, are Artificial Intelligence models designed to process and generate human language. They can be used for applications such as question answering, text generation, summarization, information extraction, coding assistance, and intelligent conversational systems.

However, general-purpose language models may not always have access to specific information required for a particular organization, domain, or collection of documents. This creates a need for systems that can combine the language-processing capabilities of an LLM with an external knowledge source.

The project undertaken during the internship focused on understanding and developing the concept of an **LLM-Based Intelligent Knowledge Retrieval System**.

The system is designed around the idea of providing an LLM with relevant information from a predefined knowledge base when answering user queries. Instead of depending entirely on the information stored within the model, the system can retrieve relevant information from its available knowledge source and use it as context for generating a response.

The internship was conducted at the **Defence Research and Development Organisation (DRDO)** in the **Cybersecurity** domain. The internship provided practical exposure to Artificial Intelligence, Large Language Models, AI agents, knowledge retrieval, and supporting technologies.

The project also helped in understanding the relationship between **data, knowledge bases, information retrieval, AI agents, and Large Language Models**.

4.2 ORGANIZATION PROFILE

The **Defence Research and Development Organisation (DRDO)** is India's premier research and development organization working in the field of defence technologies.

DRDO conducts research and development activities across a wide range of scientific and engineering disciplines. Its work encompasses various technological areas including electronics, computer science, information technology, communications, artificial intelligence, cybersecurity, aeronautics, materials science, and other specialized fields.

The increasing dependence of modern defence and technological systems on digital infrastructure makes cybersecurity an important area of research and development.

Cybersecurity involves protecting information systems, networks, software, and digital resources against unauthorized access, misuse, disruption, and other security threats.

Artificial Intelligence is also becoming increasingly relevant to cybersecurity because AI systems can assist with information processing, automation, analysis, and intelligent decision support.

During my internship, I worked as an **Intern** in the **Cybersecurity** domain. The internship provided me with practical exposure to Artificial Intelligence and Large Language Model technologies and helped me understand their potential applications in technical environments.

Due to the nature of the organization and the confidentiality requirements associated with the internship, confidential or restricted organizational information has not been included in this report.

4.3 PROBLEM STATEMENT

Large organizations and technical environments often deal with large collections of information and documentation. Finding relevant information manually from such collections can be time-consuming and inefficient.

Large Language Models provide powerful natural-language processing capabilities, but a general-purpose LLM may not have direct access to information contained within a particular private or specialized collection of documents.

Therefore, the project focuses on the following problem:

How can a Large Language Model be integrated with an external knowledge base so that relevant information can be retrieved and used to generate meaningful responses to user queries?

The proposed system addresses this problem through an intelligent knowledge retrieval workflow.

The basic concept can be represented as:

Knowledge Base → Information Retrieval → Relevant Context → LLM → Generated Response

A user submits a natural-language query. The system identifies relevant information from the available knowledge source and provides the retrieved information to the LLM. The LLM then uses the available context to generate a natural-language response.

This approach can make an LLM-based application more useful for specialized information domains.

4.4 PROJECT OBJECTIVES

The primary objectives of the project were:

1. To understand the fundamentals of Artificial Intelligence and Large Language Models (LLMs).
 2. To study the role of AI agents in developing intelligent systems.
 3. To understand the concept of external knowledge bases and intelligent information retrieval.
 4. To explore how relevant information can be retrieved from documents and provided as context to an LLM.
 5. To understand the development and working of an LLM-based question-answering system.
 6. To gain practical exposure to Python, SQL, C++, Virtual Machines, and OpenCode.
 7. To strengthen programming, debugging, problem-solving, and system development skills.
 8. To understand the importance of data security, confidentiality, and responsible information handling in cybersecurity environments.
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4.5 SCOPE OF THE PROJECT

The scope of the project focuses on developing an understanding of intelligent information retrieval using Large Language Models and an external knowledge source.

The system can be used to retrieve relevant information from a predefined collection of documents and provide the information to an LLM for generating a natural-language response.

The major areas covered by the project include:

- Large Language Models
- Artificial Intelligence
- AI agents
- Knowledge bases
- Information retrieval
- Natural-language queries
- Context-based response generation
- Document processing
- AI-assisted development

The system can potentially be adapted for specialized document collections where users need to obtain information using natural-language questions.

Potential application areas include:

- Technical documentation
- Research documentation
- Enterprise knowledge systems
- Cybersecurity information
- Educational resources
- Organizational documentation

The scope of the internship project was primarily focused on understanding and implementing the core concepts rather than developing a complete production-level enterprise application.

4.6 TECHNOLOGIES AND TOOLS USED

4.6.1 Python

Python was one of the primary programming languages used during the internship.

Python is widely used in Artificial Intelligence, machine learning, automation, data processing, and software development due to its simple syntax and extensive library ecosystem.

Python was used for programming and supporting various components of the AI and knowledge-retrieval workflow.

4.6.2 Large Language Models

Large Language Models were the primary AI technology explored during the internship.

LLMs are capable of processing natural-language input and generating natural-language responses.

In the project, LLMs were studied in the context of using external knowledge to improve information retrieval and question answering.

4.6.3 AI Agents

AI agents were explored as intelligent systems capable of using AI models and available tools or information sources to perform tasks.

An AI agent can follow a workflow such as:

User Query → Processing → Information Retrieval → Reasoning → Response

The study of AI agents helped provide an understanding of how LLMs can be incorporated into larger intelligent systems.

4.6.4 SQL

SQL stands for Structured Query Language.

It is widely used for managing and querying structured data stored in relational databases.

Exposure to SQL helped strengthen understanding of:

- Data storage
- Queries

- Tables
- Filtering
- Data retrieval
- Database operations

4.6.5 Virtual Machines

Virtual Machines provide software-based computing environments that operate independently within a physical computer.

They are useful for:

- Testing software
- Creating isolated environments
- Experimentation
- Running different operating systems
- Security-related activities

Working with Virtual Machines provided exposure to controlled computing environments.

4.6.6 OpenCode

OpenCode was explored as part of the AI-oriented development workflow.

It provided exposure to AI-assisted development and interaction with AI-based development environments.

The tool helped in understanding how AI can be incorporated into software development and technical workflows.

4.6.7 C++

C++ was also used/explored during the internship.

C++ is a general-purpose programming language widely associated with systems programming and performance-oriented applications.

Working with C++ provided additional exposure to programming concepts and strengthened programming fundamentals.

4.7 SYSTEM ARCHITECTURE

The proposed LLM-based knowledge retrieval system consists of several major components.

1. Knowledge Base

The knowledge base contains the collection of documents or information available to the system.

2. Query Input

The user provides a question or information request using natural language.

3. Query Processing

The user's query is processed to determine the type of information required.

4. Information Retrieval

The system searches the available knowledge source and identifies information relevant to the query.

5. Context Formation

The retrieved information is organized into useful context.

6. Large Language Model

The LLM receives the user query along with the relevant context and generates a response.

7. Response

The final response is presented to the user.

4.8 METHODOLOGY

The methodology followed for the project can be divided into several stages.

Stage 1 — Understanding the Problem

The first stage involved understanding the limitations of relying solely on an LLM's internal knowledge and the requirement for an external knowledge source.

Stage 2 — Preparing the Knowledge Source

A collection of relevant information was organized into a knowledge base that could be accessed by the system.

Stage 3 — Query Input

The user submits a question using natural language.

For example:

“What information is available about the requested topic?”

The actual queries used during development depended on the available knowledge source.

Stage 4 — Query Processing

The system processes the query and determines what information is required to answer it.

Stage 5 — Information Retrieval

The system searches the available knowledge source for information relevant to the user's query.

Stage 6 — Context Preparation

The relevant information retrieved from the knowledge source is prepared as context for the LLM.

Stage 7 — LLM Processing

The LLM receives the query along with the relevant contextual information.

Stage 8 — Response Generation

The LLM processes the input and generates a natural-language response.

Stage 9 — Response Evaluation

The generated response can be evaluated based on relevance, clarity, and consistency with the available knowledge.

4.9 EXPECTED OUTCOMES

The project was expected to provide the following outcomes:

1. Better understanding of Artificial Intelligence and its applications.
 2. Understanding of Large Language Models and AI agents.
 3. Knowledge of external knowledge bases and information retrieval.
 4. Understanding of how external information can be provided to an LLM.
 5. Practical exposure to Python, SQL, C++, Virtual Machines, and OpenCode.
 6. Improvement in programming, debugging, and problem-solving skills.
 7. Understanding of challenges involved in developing AI-based applications.
 8. Awareness of data security, confidentiality, and responsible information handling.
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